

PART 1: PYTHON ADVANCE COURSE CONCEPTS

BASIC CONCEPTS

1. Introduction to Python (What makes Python Best of the Best ?)

2. Basic Operations in Python

3. Variables

4. Data Types

- None
- Numeric
- String
- List
- Tuple
- Set
- Dictionary
- Range

5. Operators in Python

6. Bitwise Operators (Background Mechanism)

7. Number Systems Conversions

8. Swapping Variables

9. Math functions in Python

10. User input in Python

11. Conditional Statements in Python

12. Loops

13. Loop Control Statements

INTERMEDIATE CONCEPTS

1. Arrays

2. Using Numpy in Arrays

3. Copying Array & Matrix in Python

4. Functions in Python

5. Function Arguments and Types

6. Keyworded variable Length Arguments(**kwargs)

7. Global Keyword

8. Fibonacci Series and Factorial.

9. Recursion

10. Lambda Function

11. Filter, map, reduce

12. Decorators

13. Modules

14. Special Variable (__name__)

ADVANCE CONCEPTS

1. OOPs concept in Python (class / object)

2. Constructor

3. Types of Variables and Methods

4. Inner class

5. Inheritance and Constructor in Inheritance

6. Polymorphism

- Duck Typing
- Method Overloading
- Method Overriding
- Operator Overloading

7. Abstraction

8. Encapsulation

9. Exception Handling

10. File Handling

11. Searchings and Sortings

PART 2: MACHINE LEARNING INTERNSHIP

Module 1: Python Libraries for Machine Learning

- NumPy
- Pandas
- Matplotlib
- Assignment

ASSIGNMENT INCLUDED

Module 2: Data Preprocessing

- Handling Missing Data
- Encoding Categorical Variables
- Feature Scaling (Normalization & Standardization)

ASSIGNMENT INCLUDED

Module 3: Supervised Machine Learning

A. Regression Algorithms

- 1. Linear Regression
- 2. Multiple Linear Regression
- 3. Polynomial Regression
- 4. Decision Tree Regression
- 5. Random Forest Regression

PROJECT INCLUDED

B. Classification Algorithms

- 1. Logistic Regression
- 2. Support Vector Machine (SVM)
- 3. K-Nearest Neighbors (KNN)
- 4. Naive Bayes
- 5. Decision Tree Classifier
- 6. Random Forest Classifier

PROJECT INCLUDED

Module 4: Unsupervised Machine Learning

- 1. K-Means Clustering
- 2. Hierarchical Clustering

PROJECT INCLUDED

Module 5: Deep Learning Fundamentals

- Neural Networks
- Activation Functions
- Optimizers
- Loss Functions & Backpropagation

PROJECT INCLUDED

Module 6: Convolutional Neural Networks (CNN)

- Convolution, Pooling, Flattening
- CNN Architecture in TensorFlow/Keras
- Data Augmentation
- Transfer Learning (VGG, ResNet overview)

PROJECT INCLUDED

Module 7: Recurrent Neural Networks (RNN)

- Sequential Data and Time Series
- RNN, LSTM, GRU Architectures
- Text and Sentiment Analysis

PROJECT INCLUDED

Module 8: Large Language Models (LLMs)

- • Transformers Overview (Encoder-Decoder Architecture)
- • BERT, GPT, LLAMA, Gemini Overview
- • Prompt Engineering Basics
- • RAG (Retrieval-Augmented Generation)

PROJECT INCLUDED

Module 9: Model Deployment (Intro)

- • Introduction to Model Deployment
- • Streamlit Overview
- • AWS Overview